

Complete Evidence-Based Model Answers • Q1 + Q2 + Q3 • with Active-Recall Pack
Proctored Online Examination • January 2025 Semester

SUSS • SWK501 HUMAN LIFESPAN & SOCIAL ECOLOGY

The Complete January 2025 POE Model Answers & Active-Recall Pack

Q1 (60 marks) — the Max case, analysed through Bronfenbrenner's bioecological (PPCT) model + dual-focus intervention

Q2 (20 marks) — cognition, intelligence and the emergence of wisdom in middle adulthood

Q3 (20 marks) — Baltes' SOC model and the wider successful-ageing literature in Singapore's active-ageing landscape

One exam-ready compendium: three full-distinction model answers, rapid-revision matrices, and a printable flashcard + quiz pack for active recall — grounded in longitudinal, meta-analytic and official Singapore evidence.

ORIENTATION

How to use this compendium

This single document compiles evidence-based model answers to all three compulsory questions of the SWK501 January 2025 Proctored Online Examination, followed by an active-recall pack. Each answer follows the SUSS marking hallmarks: a multidimensional, non-deterministic account of development; explicit theory with named authors; and Singapore-grounded application. Population-level findings are treated as probabilistic tendencies, never as deterministic rules for any individual. Sources are paraphrased; a consolidated reference list closes the document.

- **Sections 1–3** are the full model answers (Q1 Max; Q2 cognition/wisdom; Q3 SOC).
- **Section 4** is a one-screen rapid-revision sheet across all three questions.
- **Section 5** is the active-recall pack: covered-answer flashcards and a 14-item MCQ quiz with an answer key.
- **Section 6** lists the sources.

***Golden thread across all three answers:** development in midlife and later life is multidirectional — gains, stability and losses co-occur — and outcomes are shaped as much by ecology and structure as by the individual. Name mechanisms, not just categories; resist determinism.*

QUESTION 1 • 60 MARKS

The Case of Max — Bronfenbrenner's Bioecological (Socio-Ecological) Model

Case anchor & blueprint

Max, 14 (Secondary Two), was previously quiet, polite, academically engaged and good at sport. After rapid physical growth he became louder and more active, increasingly late or absent, stopped submitting homework, performed poorly, argued with classmates, smoked and drank with older (16-year-old) school dropouts, and was arrested for shop theft. His parents worked long hours with little time for supervision. SUSS examiners use this vignette to test whether a candidate can move **beyond the static five-ring diagram** to the mature model, in which **proximal processes are the “engines of development”** (Bronfenbrenner & Morris, 2006), and then convert that analysis into a **dual-focus intervention** that changes both Max and his ecology.

#1 marking signal: every system named must be tied to a repeated, reciprocal interaction (a proximal process) and to evidence. A ring with no mechanism is a low-yield point.

1. The Person — characteristics that steer proximal processes

Bronfenbrenner distinguished force, resource and demand characteristics; the person is **both a producer and a product** of development. Max is not a passive recipient of his environment.

- **Force** (temperament, agency): a shift from quiet compliance to active, status-seeking agency — he *selected* the older peer group, so he is not merely a victim of peer pressure.
- **Resource** (abilities, assets): prior academic effort, teacher praise and sporting competence are *recoverable* protective assets — the raw material for intervention.
- **Demand** (features that invite responses): age 14, rapid growth, male, larger and louder — changing how older adolescents and adults respond to him. Pubertal timing is associated with externalising behaviour in boys but is environmentally moderated and non-uniform.

The **dual-systems model** (Steinberg) frames adolescence as a window in which a reactive reward system outpaces still-maturing cognitive control — the person-level vulnerability that context then activates.

2. The Process — proximal processes as the engines of development

Proximal processes are progressively more complex, reciprocal interactions with people, objects and symbols that, to be developmentally effective, must occur **on a fairly regular basis over extended periods of time** (Bronfenbrenner & Morris, 2006). Disorganised or disadvantaged interactions tend to produce **dysfunction** rather than **competence**. For Max, the *daily reinforcing* interactions with older dropouts (modelling and rewarding smoking, loud behaviour, rule-breaking and theft) displace competence-building interactions with teachers, sport and family. This peer mechanism is supported by fMRI evidence that peer presence heightens adolescent reward-circuit activity (ventral striatum, orbitofrontal cortex) and subsequent risk-taking (Chein, Albert, O'Brien, Uckert & Steinberg, 2011).

3. The Context — the nested systems applied to Max

System	In Max's case	Developmental consequence
Micro	Family (low supervision), school (disengagement), older peers, void-deck/hawker-centre gathering	Loss of monitoring, competence and conventional belonging; belonging sought in the gang
Meso	The broken parent–school link : the teacher tried; the parents' hours blocked contact	A settings-to-settings channel that would catch early disengagement did not fire
Exo	Parents' long-hours, low-control workplaces (Max does not participate in them)	Indirectly reduces supervision and family–school contact — a non-blaming structural account
Macro	Singapore's rehabilitative youth-justice ethos (CYPA), meritocratic schooling, filial norms	Shapes how disengagement and offending are experienced and responded to; access to activities is unequal

System	In Max's case	Developmental consequence
Chrono	Well-functioning → puberty → disengagement → peers → substances → theft, across school terms	Timing and sequence make this a developmental formulation, not a static “delinquent” label

Leverage point: *the weakened mesosystem (parent–school–worker channel) is usually higher-yield than any single-setting fix, and the exosystem framing explains reduced monitoring without moralising about “bad parents.”*

4. Why bioecological theory is most applicable — and where it needs help

It is preferable because it integrates biological maturation, personal agency, reciprocal interaction, family constraint, school process, peer influence, neighbourhood opportunity and change over time — mapping directly onto SUSS's dual-focus mandate. Its limitation: it can become merely descriptive if systems are listed without mechanisms (a documented “misuse”; Tudge et al.), and it does not explain *why Max learned to steal*. The high-distinction move is to keep bioecology as the integrative frame and add **social-learning** / **differential-association** mechanisms (modelling, reinforcement, definitions favourable to offending, opportunity) to explain the peer process.

5. Dual-focus intervention grounded in Max's ecology and Singapore's youth-justice system

- **Diversion first (macro → micro):** shoplifting is Singapore's most common youth offence; for a minor first offence the MSF **Guidance Programme** (6-month diversionary) offers a warning in lieu of prosecution. If the group is gang-linked, the **Enhanced Streetwise Programme** is the targeted track.
- **Person + micro:** motivational interviewing (surface identity discrepancy) plus **CBT** (behaviour-chain analysis, refusal skills, consequential thinking) — CBT reduces youth recidivism when risk-matched and well-implemented (Campbell review). **Restorative, not humiliating** accountability; **avoid Scared Straight** (ineffective/harmful; Petrosino).
- **Family + meso:** FFT-informed work — reframe from “bad child/neglectful parents” to a family in a cycle; build *feasible* monitoring and a trusted adult for cover. A Singapore FFT RCT improved probation completion (88.9% vs 70.2%), though international evidence is mixed — use it as monitored, not presumed superior.
- **School + meso:** case conference; realistic attendance target with same-day follow-up; a named mentor; academic recovery in *selected* subjects first; structured return to sport/CCA; recognition for progress.
- **Peer + community:** replace (not merely remove) status and belonging; **avoid homogeneous high-risk group aggregation** (deviancy training; Dishion); use a Volunteer Probation Officer mentor and a mixed-risk group.
- **Macro / structural:** address affordability, transport and family financial stress so the plan does not assume resources the family may lack.
- **Evaluation:** baseline, week 8, week 16 and 3- / 6-month follow-up across offending, school, substance use, family functioning, peers/community and psychological functioning. Success = sustained functioning after support tapers, not 16 weeks of obedience.

6. Critical appraisal & non-deterministic caveats

- No single factor (puberty, family, peers, substances) is **the** cause; effects are modest, reciprocal and interacting.
- **No psychiatric diagnosis** can be inferred from the vignette — assess if indicated, never infer.
- The direction of effects (school disengagement ↔ delinquency ↔ substance use) cannot be established from the case.

Supervisor's advisory (Q1 pitfalls): (1) ring-listing without mechanism; (2) diagnosing Max; (3) blaming the parents — use the exosystem and locate risk in the combination of unstructured time, weak adult connection, delinquent norms and opportunity.

QUESTION 2 • 20 MARKS

Cognition, Intelligence & Wisdom in Middle Adulthood

Strategic blueprint

The question probes whether a candidate can (a) distinguish **cognitive mechanics** (fluid) from **cognitive pragmatics** (crystallized) and evidence their differing midlife trajectories; (b) deploy more than one intelligence theory (Gardner, Sternberg); and (c) argue that **wisdom is distinct from intelligence** and is not an automatic dividend of age.

1. Framing & the mechanics/pragmatics distinction

Baltes' lifespan principles anchor the answer: development is lifelong, **multidirectional**, plastic and context-embedded; midlife (~40s–early 60s) boundaries are approximations. Cattell–Horn's Gf–Gc, echoed in Baltes' mechanics (biological “hardware”: speed, working memory, novel reasoning) versus pragmatics (culture-based “software”: vocabulary, knowledge, expertise), is the analytic spine. Processing speed (Salthouse) declines gradually from the 20s–30s; crystallized knowledge is typically stable or rising through midlife.

Dimension	Cognitive mechanics (fluid, Gf)	Cognitive pragmatics (crystallized, Gc)
Nature	Speed, working memory, reasoning with novel material	Vocabulary, factual/procedural knowledge, expertise
Midlife trajectory	Gradual decline; perceptual speed and numeric ability slip earliest	Stable or increasing into the 60s
Everyday meaning	Slower on unfamiliar, time-pressured tasks	Better contextual judgement in familiar domains

2. Longitudinal evidence: the Seattle Longitudinal Study

Schaie's SLS (from 1956; ~6,000 participants; cohort-sequential) found that reliable average decrements do not appear before about age 60, becoming reliable across abilities by the mid-70s (Schaie & Willis). Middle-aged adults outperformed their own younger-adult selves on four of six abilities; verbal memory, spatial skill, inductive reasoning and vocabulary rise into the 60s–70s, while numeric ability and perceptual speed decline earlier. Rank-order stability is high (~.90). The Whitehall II study (Singh-Manoux) is a useful counterpoint, detecting small declines from ~45.

3. Beyond IQ; and the wisdom theory landscape

- **Gardner's Multiple Intelligences** reframes intelligence as a profile, legitimising midlife strengths IQ misses.
- **Sternberg's practical intelligence** rests on **tacit knowledge** — experiential know-how that accrues with age.
- **Everyday problem-solving** improves into midlife, drawing on both fluid capacity and life knowledge; expertise is domain-specific and compensatory.

Wisdom model (authors)	Core definition	Emphasis
Berlin Paradigm (Baltes & Staudinger, 2000)	Expert knowledge in the fundamental pragmatics of life	Factual + procedural knowledge; lifespan contextualism; value relativism; managing uncertainty
Three-Dimensional (Ardelt, 2003)	A person-centred integration of characteristics	Cognitive, reflective, affective/compassionate
Balance Theory (Sternberg, 1998)	Applying tacit knowledge to a common good	Balancing intra-/inter-/extra-personal interests
Wise reasoning (Grossmann)	Context-sensitive reasoning (state, not only trait)	Intellectual humility, perspective-taking; self-distancing

Does age confer wisdom? Barely: the meta-analytic age–wisdom correlation is roughly $r = .04$. Midlife can still be favourable (inverted-U on some dimensions) because experience meets improved emotion regulation. The real pathways are **generativity** (Erikson; McAdams) and **exploratory, reflective processing** (Glück; self-distancing / Solomon's paradox). Knowledge is necessary but not sufficient.

4. Singapore application & appraisal

For midlife adults, **SkillsFuture Level-Up** (support for those 40+) operationalises adult plasticity — a learner applies crystallized domain knowledge while exercising fluid reasoning in a new domain; occupational complexity supports fluid outcomes, while cardiovascular risk is linked dose-responsively to memory decline. Participation is opportunity, not proof of raised intelligence. Average trajectories conceal large individual differences; wisdom constructs differ (Berlin measures expert knowledge, Ardelt measures wise persons) and most evidence is observational.

***Supervisor's advisory (Q2 pitfalls):** avoid the “universal decline” or single-“peak” narrative; do not equate crystallized knowledge with wisdom; do not treat SkillsFuture participation as proof of cognitive gain.*

QUESTION 3 • 20 MARKS

SOC & Successful Ageing in Singapore's Active-Ageing Landscape

Strategic blueprint

Explain Baltes and Baltes' **Selective Optimisation with Compensation (SOC)** meta-model and evaluate its contribution to successful ageing, with Singapore examples — positioning SOC within the wider literature and joining individual strategy to structural supports.

1. Foundations: SOC

SOC (Baltes & Baltes, 1990) is a **process meta-model**: across life, and especially as losses accumulate, people manage a shifting gain–loss balance by coordinating three interdependent processes. Baltes' exemplar is Rubinstein, who in old age played fewer pieces (selection), practised them more (optimisation), and slowed before fast passages so they sounded faster by contrast (compensation).

Process	Definition	Everyday example	Singapore / AAC lever
Selection (elective & loss-based)	Prioritising meaningful goals; revising after loss	Focus on grandchildren + one hobby	AAC helps set one meaningful goal
Optimisation	Investing and practising resources for chosen goals	Exercise, rehearsal, training	Move It Feel Strong (HPB); National Silver Academy
Compensation	Alternative means when a prior method fails	Hearing aids, reminders, a buddy	AAC assistive aids, befriending, transport

The processes are interdependent: selection without optimisation yields intention without capacity; optimisation without compensation fails once losses make the old strategy ineffective; compensation without meaningful selection maintains activity but not well-being.

2. How SOC contributes to successful ageing

- Goal clarity and reduced overload preserve motivation and a sense of control.
- Maintenance of competence through practice and resource investment; adaptation to loss keeps valued roles in altered form.
- Identity and autonomy are preserved even as a role's form changes.
- Berlin Aging Study: greater self-reported SOC use is associated with higher scores on successful-ageing indicators, and is especially valuable when resources are constrained (Freund & Baltes).

3. SOC within the wider successful-ageing field

Model (authors)	Type	Core claim	Key limitation
Activity theory (Havighurst, 1961)	Psychosocial	Sustaining activity/roles supports satisfaction	Ignores structural constraints
Rowe & Kahn (1987/1997)	Biomedical / outcome	Avoid disease + high function + engagement	Excludes disabled elders; culture-blind
SOC (Baltes & Baltes, 1990)	Process	Maximise gains, minimise losses via S/O/C	Correlational, self-report; resource-dependent
SST (Carstensen, 1992)	Motivational	Shrinking horizons → emotion-meaningful goals; positivity	Weaker on how beliefs change
Continuity (Atchley, 1989)	Psychosocial	Adapt using familiar strategies/identities	Can normalise decline
Healthy Ageing (WHO, 2015+)	Functional	Maintain functional ability & intrinsic capacity	Implementation-dependent

Integrative move: Kahn judged SOC complementary to Rowe & Kahn — the latter names the outcome, SOC names the process. Pair a process model (SOC) with a functional/ecological model (WHO).

4. Singapore's active-ageing ecology (current)

Singapore is projected super-aged in 2026, with one in four citizens aged 65+ by 2030. The **2023 Action Plan for Successful Ageing** is organised around **care, contribution and connectedness**; its vehicle, **Age Well SG** (announced Nov 2023; S\$3.5 billion over a decade; MOH/MND/MOT), anchors ageing in the community.

- **Active Ageing Centres** expanded from 154 (2023) toward 220+ (over 230 islandwide, ~8 in 10 seniors with access), running the **ABC + 2S** model across five domains including a dedicated **cognitive** domain.
- **Cognitive programmes**: Move It Feel Strong (HPB), square-stepping (SportSG; cognition + fall prevention), digital literacy, and NUS Mind Science Centre's Age Well Everyday (mindful awareness, art, music reminiscence, horticulture).
- **National Silver Academy** (C3A) for lifelong learning; **Community Care Apartments** pairing housing with social activity.

5. Proposed SOC-structured AAC activity

“My Neighbourhood Digital Story-Walk” — an 8-week AAC programme in which each senior creates a three-minute digital story about a meaningful place or memory and shares it at an intergenerational showcase, braiding cognitive stimulation, gentle activity and social connection:

- **Selection** — choose one valued story goal and a feasible mode (outdoor/indoor walk, or a seated project with existing photos); elective, becoming loss-based when revised for fatigue or mobility limits.
- **Optimisation** — weekly practice of pacing/balance, photography or audio, sequencing, and rehearsal with feedback, in manageable steps.
- **Compensation** — large-print cards, voice-to-text, tripods, hearing support, templates, a buddy, transport, and existing photos where walking is not feasible — preserving the goal even when the method changes.

Pathway to successful ageing: meaning and ownership → skills built → participation despite functional difference → self-efficacy and identity continuity → connection and generativity — aligning with care, contribution and connectedness. Success is not full independence: completing a valued story with a buddy and voice recording demonstrates compensation as clearly as unaided work.

6. Critical appraisal & structural caveats

- Much SOC evidence is **correlational and self-reported**; association is not proof of cause.
- Selection can be maladaptive (avoidable exclusion), compensation unavailable (unaffordable aids), optimisation overextended (unattainable goals).
- SOC must be joined to a **social-ecological account**: accessible buildings, affordable care, income, transport, digital inclusion.
- Acknowledge critiques of Rowe & Kahn (individualistic, culture-blind) and of “successful ageing” as neoliberal responsibilisation; Asian/holistic and life-course perspectives widen the frame.

***Supervisor's advisory (Q3 pitfalls):** do not reduce SOC to “staying active”; name the three interdependent processes; do not make the individual solely responsible — tie SOC to structural supports and Age Well SG.*

SECTION 4 • RAPID REVISION

One-screen recall across all three questions

Q1 — Max / Bronfenbrenner

- **PPCT**: Process (proximal processes = engine), Person (force/resource/demand), Context (micro–macro), Time (chrono).
- **Systems**: micro (family/school/peers/neighbourhood); meso (broken parent–school link); exo (parents' workplaces); macro (SG rehabilitative CYPA ethos); chrono (term-by-term cascade).
- **Integrate** social learning to explain the peer/theft mechanism; **intervene** dual-focus + Guidance Programme diversion; **avoid** Scared Straight and deviancy-training groups.

Q2 — Cognition / Intelligence / Wisdom

- Mechanics (fluid) decline gradually; pragmatics (crystallized) stable/rising in midlife (Cattell-Horn; Salthouse).
- SLS: no reliable decrement before ~60; several abilities peak in midlife; Whitehall II hints at decline from ~45.
- Wisdom: Berlin (expert pragmatics of life), Ardelt (cognitive/reflective/affective), Sternberg (balance); age–wisdom $r \approx .04$; generativity + reflection are the pathways.

Q3 — SOC / Successful ageing

- SOC = Selection (elective/loss-based) + Optimisation + Compensation; interdependent; Rubinstein exemplar.
- Rivals/complements: Rowe & Kahn (outcome), Carstensen SST, Atchley continuity, WHO functional — SOC is the process model.
- Singapore: super-aged 2026; 2023 Action Plan (care/contribution/connectedness); Age Well SG (S\$3.5b); AACs 154→230+ (ABC+2S, cognitive domain).

SECTION 5 • ACTIVE RECALL

Flashcards — cover the right column, recall aloud, then check

Q1 — Max / Bronfenbrenner

CUE (recall aloud)	MODEL RECALL (check yourself)
Name the four PPCT components and the “engine.”	Process (proximal processes = engine of development), Person, Context, Time (Bronfenbrenner & Morris, 2006).
Define proximal processes + the two conditions for effectiveness.	Progressively complex reciprocal interactions with people/objects/symbols; must occur regularly, over extended periods.
Classify Max's person characteristics.	Force = quiet → agentic/status-seeking; Resource = prior academic + sporting competence (recoverable); Demand = age 14, rapid growth, male, larger/louder.
Which system is the key <i>meso</i> failure?	The broken parent–school link (teacher tried; parents' work hours blocked contact).
What is the <i>exo</i> influence and why cite it?	Parents' long-hours, low-control workplaces reduce supervision — explains reduced monitoring without blaming the parents.
Why add social learning to bioecology?	Bioecology integrates systems + time but not <i>why</i> Max learned to steal; social learning supplies modelling, reinforcement, offending-favourable definitions, opportunity.
Which SG diversion fits Max, with what outcome?	Guidance Programme (6-month diversionary) → warning in lieu of prosecution; Enhanced Streetwise if gang-linked.
Two interventions to AVOID and why.	Scared Straight (ineffective/harmful; Petrosino); homogeneous high-risk groups (deviancy training; Dishion).
Top exam pitfall here.	Ring-listing without a mechanism; also: no diagnosis from a vignette; don't blame the parents.

Q2 — Cognition / Intelligence / Wisdom

CUE (recall aloud)	MODEL RECALL (check yourself)
Mechanics vs pragmatics?	Mechanics = fluid (speed, WM, novel reasoning) — gradual decline; pragmatics = crystallized (knowledge, vocabulary, expertise) — stable/rising in midlife.
SLS headline on timing of decline?	No reliable average decrement before ~60; reliable across abilities by ~74 (Schaie & Willis); several abilities peak in midlife.
Which abilities decline earliest?	Perceptual speed and numeric ability.
Study that dissents from “no decline before 60”?	Whitehall II (Singh-Manoux) — small reliable declines from ~45.
Contrast the three main wisdom models.	Berlin (expert knowledge in the fundamental pragmatics of life; 5 criteria); Ardelt 3D (cognitive/reflective/affective); Sternberg balance (tacit knowledge → common good).
Age–wisdom correlation + implication?	$r \approx .04$ (trivial); age doesn't confer wisdom — generativity + reflective/exploratory processing are the pathways.
SG midlife-learning policy to cite?	SkillsFuture Level-Up (support for 40+); crystallized + fluid interplay; participation is opportunity, not proof of raised IQ.

Q3 — SOC / Successful ageing

CUE (recall aloud)	MODEL RECALL (check yourself)
Define S, O and C.	Selection (elective/loss-based goal prioritising); Optimisation (invest/practise resources); Compensation (alternative means when old ones fail).

CUE (recall aloud)	MODEL RECALL (check yourself)
Baltes' Rubinstein exemplar?	Fewer pieces (S), more practice (O), ritardando before fast passages (C).
SOC vs Rowe & Kahn?	Rowe & Kahn = outcome/biomedical (avoid disease + high function + engagement); SOC = process; Kahn called them complementary.
Carstensen's theory + core claim?	Socioemotional selectivity — shrinking time horizons shift goals toward emotionally meaningful ones; a positivity effect.
Singapore ageing headline facts?	Super-aged by 2026; 2023 Action Plan (care/contribution/connectedness); Age Well SG (S\$3.5b); AACs 154→230+, ABC+2S, cognitive domain.
Biggest SOC caveat?	Evidence largely correlational/self-report; structural resources determine feasibility — avoid individual-blame framing.

SECTION 5 (CONT.) • SELF-TEST

14-item MCQ quiz — answers & rationale on the next page

Q1. In Bronfenbrenner's mature bioecological model, the "engine of development" is:

- (a) the macrosystem
- (b) proximal processes
- (c) the chronosystem
- (d) person demand characteristics

Q2. The broken parent–school connection in Max's case is a problem in the:

- (a) microsystem
- (b) exosystem
- (c) mesosystem
- (d) macrosystem

Q3. Max's parents' long hours in low-control jobs are an influence at the:

- (a) mesosystem
- (b) exosystem
- (c) chronosystem
- (d) microsystem

Q4. Which best explains the peer mechanism in Max's offending?

- (a) bioecological theory alone
- (b) social learning integrated with bioecology
- (c) attachment theory alone
- (d) labelling theory alone

Q5. Which intervention should be AVOIDED for Max on the evidence?

- (a) risk-matched CBT
- (b) FFT-informed family work
- (c) a Scared Straight prison visit
- (d) a named school mentor

Q6. Per the Seattle Longitudinal Study, reliable average decrements generally do not appear before about age:

- (a) 45
- (b) 50
- (c) 60
- (d) 70

Q7. Crystallized intelligence in midlife typically:

- (a) declines sharply
- (b) remains stable or increases
- (c) equals processing speed
- (d) disappears

Q8. The Berlin Wisdom Paradigm defines wisdom as:

- (a) high IQ
- (b) expert knowledge in the fundamental pragmatics of life
- (c) emotional regulation only
- (d) tacit knowledge for the common good

Q9. The approximate meta-analytic age–wisdom correlation is:

- (a) .04
- (b) .30
- (c) .55
- (d) .80

Q10. Ardelt's three-dimensional model adds which dimension the Berlin paradigm underweights?

- (a) factual knowledge

- (b) procedural knowledge
- (c) affective/compassionate
- (d) value relativism

Q11. In SOC, revising a goal because of illness or bereavement is:

- (a) elective selection
- (b) loss-based selection
- (c) optimisation
- (d) compensation

Q12. Using a hearing aid to keep participating in conversation is:

- (a) selection
- (b) optimisation
- (c) compensation
- (d) disengagement

Q13. Rowe and Kahn's model is best described as:

- (a) a process model
- (b) an outcome/biomedical model
- (c) a motivational theory
- (d) a continuity theory

Q14. Singapore's 2023 Action Plan for Successful Ageing is organised around:

- (a) care, contribution, connectedness
- (b) selection, optimisation, compensation
- (c) health, wealth, housing
- (d) prevention, rehabilitation, reintegration

SECTION 5 (CONT.) • ANSWER KEY

Quiz answers with one-line rationale

Q	Ans	Why
1	b	Proximal processes are the engines of development; the systems are the stage.
2	c	Meso = interactions between two microsystems (family and school).
3	b	Exo = a setting affecting Max without his direct participation.
4	b	Bioecology frames it; social learning supplies the peer mechanism.
5	c	Deterrence/Scared Straight visits are ineffective and can be harmful.
6	c	Schaie & Willis: no reliable decrement before ~60 (reliable by ~74).
7	b	Crystallized knowledge is stable or rises through midlife.
8	b	Baltes & Staudinger's definition; (d) is Sternberg's balance theory.
9	a	The age–wisdom correlation is trivial (~.04).
10	c	Ardelt adds a reflective and affective/compassionate emphasis.
11	b	Loss-based selection = revising/abandoning goals after loss.
12	c	Compensation = alternative means when the old method fails.
13	b	Rowe & Kahn is an outcome-oriented biomedical model.
14	a	Care, contribution and connectedness.

Scoring: 12–14 = exam-ready; 9–11 = solid, revisit weak items; ≤ 8 = re-read the relevant model answer section before re-testing. Retrieval practice beats re-reading — cover the flashcard answers and say them aloud.

SECTION 6 • SOURCES

Selected references (paraphrased; author–date)

Q1 — Bronfenbrenner, adolescence & intervention

- Bronfenbrenner, U., & Morris, P. A. (2006). The bioecological model of human development. In *Handbook of Child Psychology* (6th ed., Vol. 1, pp. 793–828). Wiley.
- Bronfenbrenner, U., & Evans, G. W. (2000). Developmental science in the 21st century. *Social Development*, 9(1), 115–125.
- Tudge, J. R. H., et al. (2009). Uses and misuses of Bronfenbrenner's bioecological theory. *Journal of Family Theory & Review*, 1(4), 198–210.
- Chein, J., Albert, D., O'Brien, L., Uckert, K., & Steinberg, L. (2011). Peers increase adolescent risk taking by enhancing activity in the brain's reward circuitry. *Developmental Science*, 14(2), F1–F10.
- Hoeve, M., et al. (2009). The relationship between parenting and delinquency: a meta-analysis. *Journal of Abnormal Child Psychology*, 37(6).
- Henry, K. L., Knight, K. E., & Thornberry, T. P. (2012). School disengagement as a predictor of dropout, delinquency, and problem substance use. *Journal of Youth and Adolescence*, 41(2), 156–166.
- Lipsey, M. W., et al. Effects of cognitive-behavioural programmes for offenders (Campbell Collaboration).
- Petrosino, A., et al. — awareness / “Scared Straight” programmes meta-analysis; Dishion, T. J., et al. — iatrogenic effects of grouping antisocial youth.
- Functional Family Therapy RCT with youth offenders in Singapore (probation completion 88.9% vs 70.2%).
- Ministry of Social and Family Development (2024). *Supporting Youth Rehabilitation Trends Report*; Guidance & Enhanced Streetwise Programmes; Children and Young Persons Act amendments (eff. 1 Jan 2025).

Q2 — Cognition, intelligence & wisdom

- Schaie, K. W. (1996; 2005). *Intellectual Development in Adulthood: The Seattle Longitudinal Study*. Cambridge / Oxford University Press.
- Schaie, K. W., & Willis, S. L. (2010). The Seattle Longitudinal Study. *ISSBD Bulletin*, 57(1), 24–29.
- Salthouse, T. A. (2004; 2012). Processing-speed theory; consequences of age-related cognitive declines. *Annual Review of Psychology*, 63.
- Horn, J. L., & Cattell, R. B. (1967). Age differences in fluid and crystallized intelligence.
- Singh-Manoux, A., et al. (2012). Timing of onset of cognitive decline (Whitehall II). *BMJ*, 344, d7622.
- Gardner, H. (1983/1999). *Frames of Mind / Intelligence Reframed*. Sternberg, R. J. (1998). A balance theory of wisdom. *Review of General Psychology*, 2, 347–365.
- Baltes, P. B., & Staudinger, U. M. (2000). Wisdom: a metaheuristic to orchestrate mind and virtue. *American Psychologist*, 55, 122–136.
- Ardel, M. (2003). Empirical assessment of a three-dimensional wisdom scale. *Research on Aging*, 25, 275–324.
- Glück, J., et al. (2013). How to measure wisdom. *Frontiers in Psychology*, 4, 405; Grossmann, I., et al. — wise reasoning & self-distancing.
- Meta-analytic age–wisdom review (age–wisdom $r \approx .04$); Erikson (1963) & McAdams — generativity.

Q3 — Successful ageing, SOC & Singapore policy

- Baltes, P. B., & Baltes, M. M. (1990). Psychological perspectives on successful aging: the model of selective optimization with compensation. In *Successful Aging* (pp. 1–34). Cambridge University Press.
- Baltes, P. B. (1997). SOC as foundation of developmental theory. *American Psychologist*, 52, 366–380.
- Freund, A. M., & Baltes, P. B. (1998). Selection, optimization and compensation as strategies of life management. *Psychology and Aging*.
- Rowe, J. W., & Kahn, R. L. (1987; 1997). *Science / The Gerontologist*, 37, 433–440.
- Carstensen, L. L. (1992; 1999). Socioemotional selectivity theory.
- Atchley, R. C. (1989). A continuity theory of normal aging. *The Gerontologist*, 29, 183–190; Havighurst, R. J. (1961). Successful aging.
- World Health Organization (2015). *World Report on Ageing and Health*.
- Ministry of Health (2023). *2023 Action Plan for Successful Ageing* (care, contribution, connectedness).
- MOH / MND / MOT (2023–2024). *Age Well SG* (\$3.5 billion over a decade); AAC network expansion (154 → 220+/230+; ABC + 2S; five domains).
- HPB / SportSG — Move It Feel Strong; square-stepping. NUS Mind Science Centre — Age Well Everyday. Council for Third Age — National Silver Academy. SkillsFuture Singapore — Level-Up (40+).

Study aid prepared for revision. Empirical associations are population-level and probabilistic; apply them as contributing mechanisms, not deterministic rules for any individual. Verify programme figures (AAC counts, budgets, policy dates) against current MSF / MOH / AIC / Age Well SG publications before submission, as these are updated periodically.